



About Arcube:

Some more information into Arcube's product, we are a post-flight ancillary sales platform. Once passengers get off a flight, we send an email or other form of communication to each passenger with personalised benefits. They can swap the loyalty points they just earned for the flight, into ancillaries or benefits that they can reserve for a future flight with that same airline.

Next time the customer is in the market for a flight, the fact they have a pre-reserved benefit with one airline means they are more likely to return to fly with them over competitors given they are "locked-in" to the airline's loyalty - without having to fly a certain number of times to get tier benefits or enough points to achieve a free flight.

In order to create personalised recommendations, we use the passenger's past travel history, miles earning/spending behaviour and our own insights from the travel industry such as road traffic data and security queue wait times.

The Task:

Develop a simplified version of a URL shortening service. This service should allow users to input a long URL and receive a shortened version, similar to what bit.ly do. The shortened URL should redirect to the original URL when accessed.

Requirements:

1. Backend Development

- Create a RESTful API with the following endpoints:
 - POST /shorten: Accepts a long URL and returns a shortened URL.
 - GET /{shortened_id}: Redirects to the original URL corresponding to the shortened ID.
- Use a database to store the mappings between long URLs and their shortened versions.
- Implement validation to ensure that the input URL is valid.

2. Frontend Development

- Develop a simple web interface where users can:
 - Input a long URL.
 - Receive and display the shortened URL.
- Ensure the frontend communicates with the backend API.



3. Testing

- Write unit tests for the backend logic.
- Write end-to-end tests for the API endpoints.
- Ensure the application handles edge cases (e.g., invalid URLs, non-existent shortened URLs)

4. Documentation

- Document the API endpoints and their usage.
- Provide a brief overview of the project structure and setup instructions.

5. Submission

- Create a Git repo and add your code to it.
- Ensure the Git repository is public in order for us to view it.
- Complete this Google Form to send us the repo link **here**.
- Please deploy on an hosting platform of your choice e.g. AWS Amplify, firebase, netlify, vercel, etc. and add the link to the deployed site in your submission form.

6. Time limit

- You have 3 days from when we send you the task to complete and send us the link to the git repo.

Technical Requirements

- **Backend:** Node and Express / Nest js
- **Frontend:** React js / Next js
- **Database:** MongoDB
- **Testing Framework:** Use appropriate testing frameworks (e.g. Jest, Mocha, Chai, etc.).

You will be assessed on code quality and creativity. We highly recommend that you add something unique or interesting to the existing requirements; to get a good score you will be expected to do more than the basic requirements. Frontend design and features is not a priority, it can be simple and practical.

Once complete, please fill in the Google Form in the “Submission” step, and the Arcube team will review your application and get back to you on the next steps of the hiring process.